16 An optical telescope received yellow light of wavelength 6×10^{-7} m from two yellow stars which are 1×10^{11} m apart and can just be resolved by the telescope.

The two stars are 6×10^{16} m from the telescope.

Which statement is **incorrect**?

A The aperture of the telescope has a diameter of about 0.4 m.

B The resolving power of the telescope depends on both the wavelength of light received and the aperture diameter of the telescope.

C The telescope receiving red light of wavelength 7×10^{-7} m can be used to resolve two red stars of the same separation and distance from the telescope as the two yellow stars.

D The telescope receiving yellow light of wavelength 6×10^{-7} m can be used to resolve another pair of yellow stars that are separated by 3×10^{-6} radians.

Ans: **C**

$$s = r\theta, \quad 1\times10^{11} = \left(6\times10^{16}\right)\theta$$

Option A is correct: $\theta = 1.7\times10^{-6}$ rad

$$\theta = \frac{\lambda}{b}, \quad 1.7\times10^{-6} = \frac{6\times10^{-7}}{b}, \quad b = 0.35 \approx 0.4 \text{ m}$$

Option B is correct by Rayleigh's formula.

Option C is incorrect as red light has a longer wavelength, so the telescope's minimum resolvable angle becomes larger than the actual angular separation between the two stars which was at the same distance apart and from the telescope.

Option D is correct as 3×10^{-6} rad is larger than the minimum resolvable angle of 1.7×10^{-6} rad.

17 Which is **not** an assumption of the kinetic theory of gases?